■# FLCR-31 - Gauge Groups Translated Into LCR

**Series:** Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.


# Abstract

This paper formalizes gauge-group translation from mature LCR-native results. The operative object is gauge translation. The core result is that gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited. The paper also defines how this result routes forward: this paper opens the Standard Model translation band and depends on FLCR-01 through FLCR-14. Its main residue is explicit: measured gauge coupling values and physical identity require citation/calibration lanes.



# Keywords

gauge translation; LCR; receipt; claim lane; normal form.


# External Reader Orientation

An outside reviewer should read this paper as a translation and comparator paper. Its local object is **Gauge Groups Translated Into LCR**. The paper's immediate contribution is: Translates gauge-group language into mature LCR-native terms.

The nearest external anchors are standard mathematical physics definitions, cited Standard Model targets, calibrated constants or data where available, and the LCR-native papers that provide the source-side object. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It may close the external target definition and the internal translation grammar, but it does not claim measured physical identity unless a calibration/comparator row is present.

# Reviewer Compression

**Core object.** Gauge Groups Translated Into LCR.

**Primary result.** This paper contributes a controlled mapping between a cited external target and a cited LCR-native source object.

**Primary non-result.** This paper does not claim measured physical identity unless the comparator/calibration row is attached.

**Review strategy.** Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

# Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

**Recommendation.** major revision, technically promising.

**Review score vector.** orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, external_target_boundary=2.

**Formal claim rows reviewed.** 3.

## Reviewer Strength Finding

The manuscript correctly treats Standard Model language as an external target surface and keeps translation claims separate from measured physical identity.

## Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add a comparator table mapping each external target object to the exact LCR source object, invariant, calibration status, and falsifier.

### Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.
- State scale, scheme, and units for any physics-facing numerical comparator.

### Acceptance Blockers

- Measured physical identity cannot be accepted without comparator/calibration rows.
- The mapping from external target to LCR source must be explicit enough that another reviewer could reject or reproduce it.

## 1. Contribution And Scope

- Defines gauge translation as a first-class FLCR object.
- States the local result: gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited.
- Routes downstream use through claim lanes rather than inherited prose: this paper opens the Standard Model translation band and depends on FLCR-01 through FLCR-14.
- Preserves the residue boundary: measured gauge coupling values and physical identity require citation/calibration lanes.

This paper belongs to the Standard Model translation band. Standard Model language names an external target surface, not an automatic LCR identity. A translation claim is admissible only when the cited standard object, the LCR-native source object, the mapping rule, the evidence lane, and the falsifier or calibration boundary are all visible.

## 2. Source Routing

Primary routed sources:

- `13_Standard_Model_Quark_Face_Transport.md`
- `supplements/SM_BRIDGE_SUPPLEMENT.md`

Cross-corpus evidence classes:

- `CAM_CLAIM_100_SCORE_AUDIT.md`
- `NSIT_TOOL_CLOSURE_MATRIX.md`
- `NSIT_REASONED_CLOSURE_PASS.md`
- `PAPER_REWORKS_CRYSTAL_PROJECTION.json`
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

## 3. Definitions

- **Gauge translation.** A mapping from LCR-native operations into gauge-theoretic language.
- **Dependency citation.** The required reference back to the LCR-native paper proving the source object.
- **Receipt boundary.** The exact scope in which the paper's result can be replayed or consumed.
- **Promotion route.** The evidence path required before a stronger downstream claim can use this result.

# 4. Formal Claims

| Claim | Lane | Statement |
|---|---|---|
| Theorem 31.1 | `standard_theorem_citation_bound_result` | gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited |
| Proposition 31.2 | `normal_form_result` | gauge translation can be imported by later papers only through its declared source, receipt, and lane. |
| Protocol 31.3 | `falsifier_or_open_obligation` | measured gauge coupling values and physical identity require citation/calibration lanes |

# Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

| Formal row | Type | Claim in review terms | Evidence required | Boundary | Next review action |
|---|---|---|---|---|---|
| Theorem 31.1 | external target or comparator | gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited | named external theorem, definition layer, citation target, or imported standard formalism | imports the external object as-is; does not silently reprove or redefine it | attach final citation metadata and mapping rule |
| Proposition 31.2 | translation grammar | gauge translation can be imported by later papers only through its declared source, receipt, and lane | definitions, normal-form construction, conversion rule, and downstream-use boundary | internal formal coherence; no measured physical identity without a separate calibration row | check that the normal form is named and the conversion rule is explicit |
| Protocol 31.3 | obligation/falsifier | measured gauge coupling values and physical identity require citation/calibration lanes | explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition | does not negate satisfied lower-level rows | name the next binding action and owner surface |

# Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

| Claim type | What this paper may claim | Acceptance test | What is not claimed by that row |
|---|---|---|---|
| Formal-system result | `FLCR-31` defines or uses **Gauge Groups Translated Into LCR** as a typed FLCR object with declared inputs, operations, outputs, and residue. | Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed. | External physical truth, measured prediction, or identity with a standard theory. |
| Computational or receipt-bound result | Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain. | A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope. | Completeness outside the enumerated domain or physical correctness outside the tested comparator. |
| Imported theorem or external definition | The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present. | The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit. | A new proof of the imported theorem or a hidden change in the imported object's meaning. |
| Interpretive bridge or analogy | Analog, workbook, visual, or translation language may explain why the construction is useful. | The analogy preserves the relevant structure and does not promote itself into a theorem. | That the analogy alone proves the formal, computational, or physical claim. |
| Physical or calibration-facing claim | Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria. | A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached. | A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier. |
| Open obligation or falsifier | A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel. | The next binding action and the reason the stronger claim is not closed are stated. | That the base formal result is false merely because a stronger downstream claim remains unfinished. |

# 5. Paper-Specific Development

1. Identify the local object and its assigned sources for FLCR-31.
2. Separate what is internally receipt-bound from what is citation-bound, CAM-bound, calibration-bound, or still an obligation.

3. State the strongest admissible claim in its current lane.
4. Export only the lane-safe result to downstream papers and preserve the residue.

### Proof Sketch

The proof strategy for FLCR-31 is a typed construction argument. The paper names gauge translation as the active object, binds it to the routed legacy and orbital sources, and then allows downstream use only through the declared claim lane. This does not erase stronger ambitions; it keeps them addressable as dependencies, calibration tasks, or falsifiers.

## 6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is `Gauge Groups Translated Into LCR`. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

## 7. Evidence And Receipts

| Evidence source | What it contributes |
|---|---|
| Legacy paper body | Primary formal and source-integration material assigned by the series map. |
| Internal and pyramid supplements | Cross-paper reasoning windows, deferred back-application slots, and route constraints. |
| NSIT tool closure matrix | Existing verifier/tool families that can close internal or batch claims. |
| CAM/crystal and standards-window evidence | Projected memory, source routing, standards reports, and window/node databases. |

### Standard Model Definition Dependencies

This paper consumes the dedicated Standard Model Defining layer. These papers define the external Standard Model or adjacent standard-physics object before this FLCR paper translates it into LCR language.

| SMD paper | Definition surface | Required use |
|---|---|---|
| `SMD-01` | Standard Model Definition Contract And Evidence Boundary | Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim. |
| `SMD-02` | Gauge Principle, Connections, And Local Symmetry | Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim. |
| `SMD-03` | Gauge Group SU3 SU2 U1 And Representation Content | Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim. |
| `SMD-04` | Fermion Fields, Generations, Flavor, And Mixing | Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim. |
| `SMD-06` | Electroweak Interaction And Symmetry Breaking | Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim. |
| `SMD-08` | Renormalization, Effective Field Theory, And Running Parameters | Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim. |

This paper also imports SMB-01, the Standard Model Closure Bridge. SMB-01 supplies the closure-by-lane rule: rows are no longer treated as unresolved by default; they are closed when standard formalism, FLCR proof, CAM/crystal reapplication, normal-form translation, or external calibration satisfies the lane, and they remain obligations only when a required field is missing.

## Closed Rows Applied In This Paper

| Row | Closure | Evidence | Boundary |
|---|---|---|---|
| Standard Model gauge group target | closed by standard formalism | `SMD-03`, PDG source registry | Closes the external target `SU(3)_C x SU(2)_L x U(1)_Y`, not an LCR identity claim. |
| Gauge transport vocabulary | closed by standard formalism | `SMD-02` | Closes local symmetry/connection/covariant-derivative vocabulary as the external target. |
| LCR gauge-translation admissibility | closed by FLCR protocol | `FLCR-01`, `FLCR-12`, `FLCR-14`, `SMD-01` | A translation row is admissible only when the mapped LCR object, external gauge object, lane, and falsifier are present. |

The paper therefore no longer leaves gauge group translation as a generic open problem. The open part is narrower: a specific LCR object must still be mapped to a specific gauge object with representation, invariant, and calibration fields when physical identity or measured coupling values are claimed.

## Active Comparator Rows

| Comparator | External side | LCR side | Current result | Next required action |
|---|---|---|---|---|
| Gauge-group definition | `SU(3)_C x SU(2)_L x U(1)_Y` | FLCR carrier/face/atlas surfaces | closed as a citable target | Attach representation rows for each physical label. |
| Local transport | connection/covariant derivative | LCR transport and receipt paths | structurally admissible | Name the invariant preserved by the LCR transport. |
| Coupling values | PDG/evaluated couplings | none yet | calibration-bound | Add scale, scheme, value, uncertainty, and comparator. |

## Executable Evidence Bound In This Pass

This paper now binds the strongest available internal gauge-adjacent receipt instead of leaving gauge translation as an unresolved abstraction. The receipt is not used to claim a measured Standard Model identity; it is used to close the finite structural transport row that a gauge translation paper needs before it can speak precisely.

| Evidence | Path | Result imported here | Boundary preserved |
|---|---|---|---|
| Quark-face literalization receipt | `local evidence artifact` | `10/10 PASS`; three colors, S3 Weyl action, exact n=3 closure, trace-conserved transport, J3(O) face partition, and color-confinement neutral-extreme checks. | Imports structural gauge-like transport only; physical quark identity and measured couplings remain calibration-bound. |
| Quark-face transport receipt | `local evidence artifact` | S3 closes on the trace-2 triple; exact SU(3) Weyl group-ring residual is zero; six-face color/anticolor analog is internally consistent. | The receipt explicitly leaves physical quark color charge, CKM phase, and V-A weak structure as open external derivations. |
| runtime standards/window pass | `local evidence artifact` | Routes `verify_quark_face_transport_literalized.py` to `FLCR-14`, `FLCR-31`, `FLCR-32`, and `FLCR-40` as a receipt-bound internal result. | The route makes the receipt admissible for translation; it does not erase comparator or calibration fields. |

The upgraded claim is therefore: FLCR-31 can cite an internally closed finite transport surface that has the right local ingredients for gauge translation. It cannot yet claim that the whole Standard Model gauge sector has been derived until representation rows, scale-dependent couplings, and physical calibration data are attached.

## Online Source Rows Applied In This Pass

| Online source | Claim-lane effect | Boundary retained |
|---|---|---|
| `IRL-PDG-2026` | Closes the current external Standard Model review surface for gauge groups, particle listings, and evaluated parameters. | Does not prove LCR identity; it defines the external target. |
| `IRL-NIST-ALPHA-2022` | Supplies a numerical calibration target for alpha inverse: `137.035999177`, standard uncertainty `0.000000021`. | Coupling claims still require scale, scheme, and comparator. |
| `IRL-CODATA-2022-WALLET` | Supplies compact constants and unit rows for calibration tables. | Constants are not a derivation of gauge coupling from LCR. |

Online data therefore narrows the open row: FLCR-31 no longer lacks external gauge/coupling references; it lacks only the explicit LCR-to-value derivation or calibrated comparator row.

## Assertive Closure Promotions

| Row | Closure now allowed | Evidence | Residue moved out of the open row |
| --- | --- | --- | --- |
| External gauge target | closed by citation/definition | `IRL-PDG-2026`, `SMD-02`, `SMD-03` | LCR physical identity. |
| Coupling calibration target | closed as available calibration data | `IRL-NIST-ALPHA-2022`, `IRL-CODATA-2022-WALLET` | LCR derivation of the value; scale/scheme-specific comparison. |
| Finite gauge-like structural transport | closed internally | quark-face transport receipts, `FLCR-14`, `FLCR-31`, `FLCR-32` | Full gauge-sector derivation. |

The non-timid conclusion is: FLCR-31 should not call gauge translation broadly open. The open item is narrower: deriving or calibrating a specific LCR transport against a specific measured gauge quantity.

# Appendix A. Recursive Capability Reapplication Review

This review treats the newly admitted capability families as operators. The question is not only what each source says, but what it solves when the source is recursively reapplied through CAM, claim lanes, forge admission, receipt loops, and paper-to-paper routing.

## Operator Effects

| Operator | Standalone result | Recursive result | Newly resolved state | Remaining boundary |
| --- | --- | --- | --- | --- |
| REAPPLY-GLM-OBLIGATION-CLOSURE | The GLM closure matrix independently compares verifier findings to named obligations and advances 11 obligations to partial closure while identifying the untouched set. | When fed back through the claim lanes, the matrix stops the suite from carrying stale open labels where bounded verifier evidence already exists, while preserving untouched obligations as active next channels. | Obligation status promotion becomes auditable instead of rhetorical.; Papers 09, 10, 13, 16, 18, and 33 gain bounded closure support from specific verifier findings.; The suite gains a clean split between partial closure, engineering scope, and untouched physical/calibration bridges. | Partial closure is not unconditional closure.; CKM, quark/color measurement, Higgs/QFT mass calibration, GR curvature bridges, and full Moonshine remain separate dependency lanes where the matrix says they remain open. |

## Combined Recursive Effects

| Combination | Result | Solves |
| --- | --- | --- |
| COMBO-APPLIED-PHYSICS-BOUNDARY | Applied forge topology supplies the domain boundary, GLM closure supplies bounded mathematical promotions, and address horizons supply finite computational evidence. | This lets later Standard Model and physics-facing papers state strong bounded claims while separating calibration-dependent physical identity claims. |

The practical consequence is that several prior gaps are now better classified. Some are closed as routing, admission, finite-address, or executable-publication problems. The residues that remain are sharper: exhaustive source intake, physical calibration, full paper-local runner parity, external measurement, and domain-specific validation.

# Appendix B. Broad Capability Intake

This addendum admits non-obvious source material by function rather than filename. Each row states what the source now lets the paper do, while detailed receipt provenance remains in the evidence report instead of the formal claim body.

| Capability | Lane | Paper effect | Boundary |
| --- | --- | --- | --- |
| CAP-GLM-OBLIGATION-CLOSURE | `normal_form_result` | Adds a closure matrix that compares independent GLM verifier findings against open obligations and records bounded promotions instead of leaving the papers in their earlier unresolved posture. Evidence facts: 12 findings considered against 29 obligations; 11 obligations advanced to partial; 17 untouched; closed-bounded findings include GLM-F1-SM01, GLM-F2-SP002, GLM-F4-RULE30P3, GLM-F6-YM-EXT02, GLM-F7-SM04, GLM-F8-HODGE-EXT01. | Promotions are lane-bound and partial where the matrix says partial; external physical bridges remain separate calibration obligations. |

The resulting claim posture is broader but still auditable: a paper can import a capability when the evidence lane is satisfied, and any remaining gap is named as an adapter, calibration, rerun, or falsifier obligation.

# Appendix C. Implementation Intake

This addendum binds implementation work that was not fully represented in the earlier paper routing. The rows below are not pasted source text; they are evidence-lane imports that change what the paper can claim now.

| Implementation source | Lane | Claim effect | Boundary |
|---|---|---|---|
| IMPL-R30-LATTICE-THEOREM-REGISTRY | `standard_theorem_citation_bound_result` | Adds executable theorem registry rows for octonions, J3(O), chart-to-J3(O), exact n=3 SU(3) Weyl closure, Rule-30 8x8 transition, Niemeier routes, relational qubit closure, Monster/Pariah boundary, and computational ISA signatures. Verification: Targeted pytest passed: 11 Rule-30 normal-form/proof-shell tests, including address decomposition, centroid state, proof-shell dependency acyclicity, Niemeier obligation boundaries, scoped claim promotion, and full dependency closure. | Imports verifier-backed internal/formal results; transported physical interpretations remain lane-bound. |

The immediate paper effect is stronger claim posture where these implementation rows satisfy the relevant lane. Remaining caution is reserved for specific claims that still lack an adapter, comparator, calibration target, or rerun receipt.

# Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

## Closed Now

| Row | Lane | Resolved state | Remaining boundary |
|---|---|---|---|
| paper lift enumeration | `receipt_bound_internal_result` | 8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.510476. | This closes the paper-local finite lift readout, not every stronger physical interpretation. |
| closure/lift governance | `normal_form_result` | The paper's object is closed as a replayable closure/lift state with explicit open-slot preservation. | The family closure is a structural lane; measured physics still requires destination-specific calibration. |
| decade-4 tower | `receipt_bound_internal_result` | The decade tower is resolved with avg TarPit mass=0.509077 and mass sum=40.726174. | The decade total is an internal tower metric, not a standalone universal physical constant. |
| family-01 cross-lift comparison | `cam_crystal_reapplication_result` | Family tower binds FLCR-01, FLCR-11, FLCR-21, FLCR-31; avg TarPit mass=0.509490; error walls=0. | Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries. |
| gauge target and transport | `standard_theorem_citation_bound_result` | Gauge target definitions and finite gauge-like structural transport are closed. | Measured couplings require scale/scheme comparator rows. |

## Claim Posture After This Pass

`FLCR-31` should now be read as a resolved-state contribution for `Gauge Groups Translated Into LCR` at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

# 8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- `FLCR-11` through `FLCR-18` for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.

- `FLCR-28` through `FLCR-30` for CAM/crystal, corpus ribbon, and universal

normal-form intake.

- `FLCR-31` through `FLCR-40` only after the LCR-native result has been cited

and the translation claim has its own evidence lane.

# 9. Limitations And Falsifiers

- measured gauge coupling values and physical identity require citation/calibration lanes
- External calibration claims require units, datasets, citations, and reproducible data binding.
- A later translation paper may strengthen this result only by adding the missing lane evidence.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

# 10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

# Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): `13`/fold_action.

| Original slot | Family | Lift | Role | Proof form |
|---|---|---|---|---|
| `13` | fold_action | 1 | order-2 slot-3: fold the initial action into a higher-dimensional representation | fold map, coordinate atlas, and representation transport |

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

# State-Bound Proof Contract

This paper receives: state emitted by prior slot 12 and reopened at original slot 13 (Standard Model Quark Face Transport).

It must produce: formal result of original slot 13 plus the same-family lift path toward slot 23.

Semantic transition: fold the local state into a higher representation while preserving addressability.

Accepted formalism targets to bind in the journal rewrite:

- representation theory
- group actions and equivariance
- tensor or coordinate atlases
- transport maps between equivalent descriptions

| Slot | Family | Lift | Produced proof form |
|---|---|---|---|
| 13 | fold_action | 1 | fold map, coordinate atlas, and representation transport |

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

# Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

## Expansion Rule For This Slot

Use past work to strengthen representation transport: every face, gauge, atlas, or fold label must point to a preserved source object.

## Primary Evidence Inputs

| Source | Placement reason |
|---|---|
| `13_Standard_Model_Quark_Face_Transport.md` | primary assigned source for the paper's formal spine |
| `supplements/13_INTERNAL_REASONING_SUPPLEMENT.md` | paper-local reasoning supplement contributing to definitions, proof sketch, and limitations |
| `virtuous\13_Standard-Model_Quark-Face_Transport_VIRTUOUS.md` | prior enriched paper body; should be mined for mature claims and proof ordering |

## Secondary And Orbital Evidence Inputs

| Source | Placement reason |
|---|---|
| `supplements/SM_BRIDGE_SUPPLEMENT.md` | cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes |
| `supplements/RECEIPT_VERIFIER_CATALOG.md` | cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes |
| `supplements/INTERNAL_REASONING_PYRAMID_INDEX.md` | cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes |
| `supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md` | cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes |
| `supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md` | cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes |
| `supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md` | cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes |
| `CAM_CLAIM_100_SCORE_AUDIT.md` | series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body |
| `NSIT_TOOL_CLOSURE_MATRIX.md` | series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body |
| `NSIT_REASONED_CLOSURE_PASS.md` | series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body |
| `ORBITAL_DATA_CROSS_POPULATION_AUDIT.md` | series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body |
| `AGENT_CRYSTAL_WORK_HARVEST.md` | series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body |
| `PAPER_REWORKS_CRYSTAL_PROJECTION.json` | series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body |
| Additional source routes | Complete routing is maintained in the source-placement appendix |

## Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.

- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

### Paper-Specific CAM/Score Rows

| | | | | |
|---|---|---|---|---|
| 13 | 92 | internally closed, external calibration pending | CL affirmative verifier: exact SU(3) color transport, QuarkFaceForge 10/10, Paper 03 trace/face support | CKM and measured quark/color bridge dataset |

### Paper-Specific NSIT Closure Rows

| | | | | | |
|---|---|---|---|---|---|
| Paper 13 Standard-Model quark-face transport | batch_tool_unison_closure | SU(3) T4 closure, D12 trace conservation, side-flip chirality, J3(O) faces, QuarkFaceForge | Paper 13 | `quark_face_transport_literalized_receipt.json` 10/10 | Exact structural color-transport claim closes; physical-quark identity remains transport-scoped. |

### Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

## Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

| Queue | Lane | Promote now | Bind next | Residue |
|---|---|---|---|---|
| Symbolic Carrier Versus Physical Carrier | `external_calibration_result` | Promote addressable symbolic-carrier and transport claims where the paper names carrier, path, residue, or traversal rules. | Map every physical carrier phrase to a standard physics object, Hamiltonian/model, dataset, or calibration route before treating it as measured physics. | Physical identity, units, material constants, and measured transport remain calibration-bound. |
| CKM/Standard-Model Structural Transport Split | `standard_theorem_citation_bound_result` | Promote SU(3)/D12/J3 face transport as structural algebraic transport when the finite receipt is attached. | Attach quark-face transport receipt and keep CKM/PDG numerical values as data-binding requirements. | Measured CKM agreement, physical-quark identity, and parameter calibration remain data-bound. |

### Binding Discipline

Each queue item is represented as a delta:

```
local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
```

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

## Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

| Delta | Local claim at paper time | Later evidence | Revised representation | Boundary kept |
|---|---|---|---|---|
| Symbolic Vs Physical Carrier | This paper may use carrier language for addressable symbolic transport inside the LCR/CAM system. | Standard physics carrier terminology, local verifier receipts, material/plasma/transport supplements, and calibration routes. | Split symbolic carrier closure from measured physical carrier identity. | Physical identity, units, Hamiltonians, datasets, and measured constants remain external-calibration claims. |
| Ckm Sm Transport Split | This paper may claim structural face/color transport at the algebraic representation level. | SU(3)/D12/J3 receipts, quark-face transport verifier, and Standard Model/CKM data routes. | Promote structural transport while keeping numerical CKM/PDG agreement in data-binding lanes. | Physical-quark identity, measured parameters, and calibration remain citation/data-bound. |

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

# Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

| Evidence card | Score | Publication contribution | Boundary |
|---|---|---|---|
| CQE-paper-13.50: Paper 13.50 - Quark-Face Transport Claim Contract | 101 | Paper 13.50 exists to keep the most tempting overclaim visible. The proved object is strong because it is exact and finite. The physical identification becomes stronger only when it is separately derived, not when it ... | Source-backed support; promotion requires the paper-local claim lane and evidence boundary. |
| FORMAL: Paper 11 — C-Form: Theory Admission Gate Gluon | 77 | **C = the admission filter Gluon** — the trust anchor that filters external theories by Gluon mass matching. In the lattice_forge substrate, C is realized as the **admission gate** that: - Takes an external theory's G... | Source-backed support; promotion requires the paper-local claim lane and evidence boundary. |
| CQE-paper-13.25: Paper 13.25 - Toolkit for Quark-Face Transport | 77 | This support paper exposes the tools for Paper 13. It is not the proof-carrying paper. The proof is the finite algebraic closure in Paper 13; this toolkit shows how to inspect the same closure digitally and by ordinar... | Source-backed support; promotion requires the paper-local claim lane and evidence boundary. |
| CQE-paper-14: Paper 14 - GR Boundary-Repair Curvature | 77 | This paper defines the CQECMPLX substrate meaning of curvature as a boundary-repair demand. The closed result is a transport-ledger theorem: every route has a typed status, demonstrated rows carry zero repair score, n... | Source-backed support; promotion requires the paper-local claim lane and evidence boundary. |
| SUMMARY-II-Folded-Strand-Physics: Summary Paper II — Folded Strand Physics: The Gluon as Quark, Mass, Curvature, Tower, and Moonshine | 75 | This paper presents the **physics applications** of the Gluon formalism. We do not use the word "Gluon" because the fit is good. We use it because the substrate **IS** SU(3), and the physics IS the **standard model ga... | Source-backed support; promotion requires the paper-local claim lane and evidence boundary. |
| Additional evidence cards | 12 total | The complete card inventory is maintained in the archive evidence matrix. | Cards are source-discovery aids until bound to paper-local evidence. |

# Journal Apparatus

## Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

## Structured Contribution Statement

| Item | Statement |
|---|---|
| Publication ID | `FLCR-31` |
| Legacy source slot(s) | `13` |
| Ribbon role | order-2 slot-3: fold the initial action into a higher-dimensional representation |
| Proof form | fold map, coordinate atlas, and representation transport |
| Received state | state emitted by prior slot 12 and reopened at original slot 13 (Standard Model Quark Face Transport) |
| Produced state | formal result of original slot 13 plus the same-family lift path toward slot 23 |

## Claim-Evidence Table

| Claim | Lane | Evidence to attach | Boundary |
|---|---|---|---|
| Theorem 31.1 | `standard_theorem_citation_bound_result` | Receipt, source card, validator, citation, or CAM route named in the paper manifest | gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited |
| Proposition 31.2 | `normal_form_result` | Receipt, source card, validator, citation, or CAM route named in the paper manifest | gauge translation can be imported by later papers only through its declared source, receipt, and lane. |
| Protocol 31.3 | `falsifier_or_open_obligation` | Receipt, source card, validator, citation, or CAM route named in the paper manifest | measured gauge coupling values and physical identity require citation/calibration lanes |

## Figure Plan

| Figure | Purpose | Caption |
|---|---|---|
| FLCR-31-F1 | Representation fold atlas | Schematic showing how `FLCR-31` turns its received state into the produced state while preserving claim lanes and residue boundaries. |
| FLCR-31-F2 | Evidence routing map | Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript. |
| FLCR-31-F3 | Same-family lift context | Diagram placing this paper in the original `00-79` ribbon family and showing predecessor/successor slots. |

## In-Text Figure FLCR-31-F1: Representation fold atlas

```
received state
  -> Representation fold atlas
  -> formal claim lane
  -> evidence object / receipt / citation
  -> produced state
  -> remaining residue
```

## Table Plan

| Table | Purpose |
|---|---|
| FLCR-31-T1 | Face/representation correspondence table |
| FLCR-31-T2 | Claim-lane/evidence-boundary table |
| FLCR-31-T3 | Archive evidence card and source-backed upgrade table |
| FLCR-31-T4 | Workbook replay and falsifier table |

## Worked Example

Map one source object across two labels/faces while preserving its address. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

## Nomenclature And Glossary

| Term | Paper-local meaning |
| --- | --- |
| CAM | Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary. |
| atlas | Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary. |
| claim lane | Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary. |
| crystal | Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary. |
| face | Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary. |
| fold | Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary. |
| receipt | Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary. |
| representation transport | Defined by this paper as part of the `fold_action` proof role; final definition is recorded in the paper-local glossary. |

## Appendix FLCR-31-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

## Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

## Reviewer Checklist

| Check | Reviewer question |
| --- | --- |
| Claim lane | Does every strong claim declare its lane and evidence type? |
| Boundary | Does the paper preserve what it does not prove? |
| Reproducibility | Is there a receipt, verifier, source card, citation, or replay table for the claim? |
| Cross-paper import | Does the paper cite the prior FLCR/OPR slot before consuming its result? |
| Interpretation | Does the analog layer preserve the addressed state while changing labels/access paths? |

## Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

# 11. Publication Integration Addendum

## 11.1 Role In The Series

`FLCR-31` (`Gauge Groups Translated Into LCR`) occupies the `fold_action` role at lift depth `1`. The paper receives state emitted by prior slot 12 and reopened at original slot 13 (Standard Model Quark Face Transport). It produces formal result of original slot 13 plus the same-family lift path toward slot 23. The state transition is: fold the local state into a higher representation while preserving addressability.

The paper-local proof form is:

```
fold map, coordinate atlas, and representation transport
```

The integration result is a representation-transport chart with named invariants and residues. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

## 11.2 Formal Carrier

| Formal carrier | Function |
|---|---|
| representation theory | Formal carrier for the paper-local result. |
| group actions and equivariance | Formal carrier for the paper-local result. |
| tensor or coordinate atlases | Formal carrier for the paper-local result. |
| transport maps between equivalent descriptions | Formal carrier for the paper-local result. |

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

## 11.3 Claim Ledger

| Claim | Lane | Statement |
|---|---|---|
| Theorem 31.1 | `standard_theorem_citation_bound_result` | gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited |
| Proposition 31.2 | `normal_form_result` | gauge translation can be imported by later papers only through its declared source, receipt, and lane. |
| Protocol 31.3 | `falsifier_or_open_obligation` | measured gauge coupling values and physical identity require citation/calibration lanes |

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

## 11.4 Evidence Package

| Evidence class | Routed items | Status |
|---|---|---|
| Legacy sources | `13_Standard_Model_Quark_Face_Transport.md`, `supplements/SM_BRIDGE_SUPPLEMENT.md` | routed evidence |
| Evidence inputs | `CAM_CLAIM_100_SCORE_AUDIT.md`, `NSIT_TOOL_CLOSURE_MATRIX.md`, `NSIT_REASONED_CLOSURE_PASS.md`, `PAPER_REWORKS_CRYSTAL_PROJECTION.json` | routed evidence |
| CAM/crystal sources | `PAPER_REWORKS_CRYSTAL_PROJECTION.json`, `AGENT_CRYSTAL_WORK_HARVEST.json`, `runtime standards artifact`, `notebook-derived source bundle` | routed evidence |
| Standards-window inputs | `window_summary.json`, `node DBs`, `window DBs` | routed evidence |
| Receipts | external requirement | not attached in this source package |
| Validators | external requirement | not attached in this source package |

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

## 11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;

3. the receipt, enumeration, derivation, lookup, or external theorem supporting

the operation;

4. the residue or boundary left after the result;
5. the downstream import rule.

## 11.6 Figures And Tables

| Figure | Role |
| --- | --- |
| FLCR-31-F1: Representation fold atlas | Visualizes the proof object, routing, or boundary. |
| FLCR-31-F2: Evidence routing map | Visualizes the proof object, routing, or boundary. |
| FLCR-31-F3: Same-family lift context | Visualizes the proof object, routing, or boundary. |

| Table | Role |
| --- | --- |
| FLCR-31-T1: Face/representation correspondence table | Records claim lanes, evidence, sources, or falsifiers. |
| FLCR-31-T2: Claim-lane/evidence-boundary table | Records claim lanes, evidence, sources, or falsifiers. |
| FLCR-31-T3: Archive evidence card and source-backed upgrade table | Records claim lanes, evidence, sources, or falsifiers. |
| FLCR-31-T4: Workbook replay and falsifier table | Records claim lanes, evidence, sources, or falsifiers. |

## 11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.